

Universidad de las Fuerzas Armadas ESPE
Software Engineering
Advanced Web Development
Software Requirements Specification (SRS)

Project: Biconoir's Restaurant

Team: ByteClub

1. Functional Requirements

Requirement ID	FR-01
Name	Authentication & Access Control
Description	The system shall allow users to register and log in using an email and password. It must distinguish between Customer and Administrator roles upon login, granting access to specific dashboards accordingly.
Category	Security & Access
Priority	High

Requirement ID	FR-02
Name	Digital Menu Display
Description	The system shall display a digital menu where each menu item includes a specific name, selling price, detailed descriptive text, and a product image.
Category	Customer Interface
Priority	High

Requirement ID	FR-03
Name	Ingredient Customization
Description	The system shall allow customers to customize their selected menu items by removing specific base ingredients (e.g., removing cheese for allergy purposes) before adding the item to the cart.
Category	Order Management
Priority	High

Requirement ID	FR-04
Name	Virtual Shopping Cart
Description	The system shall provide a virtual shopping cart where customers can add items, modify quantities, and remove items before proceeding to checkout.
Category	Order Management
Priority	High

Requirement ID	FR-05
Name	Order Total Calculation
Description	The system shall calculate and display the total order cost. This calculation must integrate the Subtotal of the items and any applicable Transaction Fees (TF).
Category	Financial processing
Priority	High

Requirement ID	FR-06
Name	Table Reservation System
Description	The system shall allow customers to submit a table reservation specifying the date, time, and number of guests. The Administrator dashboard must display all active and scheduled reservations.
Category	Operations
Priority	High

Requirement ID	FR-07
Name	Location & Contact Display
Description	The system shall display an interactive map on the public interface, along with contact information (email and phone numbers) directly below the map module.
Category	Customer Interface
Priority	Medium

Requirement ID	FR-08
Name	Automated Inventory Deduction (FIFO)
Description	The system shall automatically deduct the required raw ingredients from the central inventory database the moment an order is completed. It must enforce a First-In, First-Out (FIFO) logic, utilizing the oldest available batches first.
Category	Backend Operations
Priority	High

Requirement ID	FR-09
Name	Inventory Batch Registration
Description	The system shall allow the Administrator to input new inventory batches, recording the unique SKU, quantity, purchase date, and the specific cost per unit.
Category	Operations
Priority	High

Requirement ID	FR-10
Name	Financial Profit Calculation & Reporting
Description	The system shall calculate the Profit/Loss (PL) for every

	completed order cycle by subtracting Ingredient Cost (IC) and Transaction Fees (TF) from the Selling Price (SP). It must generate reports displaying Total Profit-Loss (TPL) and Profit/Loss per Day (PLPD).
Category	Financial processing
Priority	High

Requirement ID	FR-11
Name	Read-Only Audit Log
Description	The system shall maintain an automated, centralized audit log of all transactions, inventory changes, and system actions. The Administrator can view this history but cannot modify or delete past records.
Category	Security & Auditing
Priority	High

2. Non-Functional Requirements

Requirement ID	NFR-01
Name	Usability & Responsiveness
Description	The web interface must be fully responsive, ensuring seamless navigation and correct display scaling on both desktop monitors and mobile devices.
Category	UI/UX
Priority	High

Requirement ID	NFR-02
Name	Data Security
Description	All user passwords must be securely hashed in the database before storage. Authentication sessions must be securely managed.
Category	Security
Priority	High

Requirement ID	NFR-03
Name	Real-Time Performance
Description	Inventory deduction and availability checks must execute in real-time to prevent customers from placing orders for items containing out-of-stock ingredients.
Category	Performance
Priority	High